

MOHAMMAD SHERIFF MEHMOOD

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Senior data scientist with 5 years of experience in healthcare, finance, and retail industries, specializing in advanced analytics, predictive modeling, and data-driven decision-making.

Skills

Programming	Python, Tensorflow, Numpy, Pandas, matplotlib, scikit-learn, Docker, LangChain, LangSmith, Django, Flask, fastAPI
Advance AI and ML	Machine learning, Deep learning, NLP, CNN, Generative AI, MLOps LLMOps, OpenCV, Yolo, ChatBot
Cloud Platform & DB	Google Cloud Platform (GCP), Amazone Web Services(AWS), MongoDB, SQL, Vector DB
Tools	Tableau, Git, Anaconda, JupyterNotebook, Visual Studio
Technical Concepts	CI/CD, Data structure and Algorithms, RAG, Finetunning
Core Competencies	Communication – Secretary of Toastmasters Club; addressed audiences at the PyData Conference (2021).

Professional Experience

Senior Data Scientist - NLP Engineer	LTIMindtree	Pune, INDIA	02/2024 - Present
- Collaborated with a 7-member team to design and implement an AI-powered customer support application. - Reduced customer churn by 40% and increased quarterly earnings by \$300K with intelligent, context-aware responses. - Engineered DAGs to optimize workflow management, resulting in a 40% improvement in processing time and allowing for an increase in project throughput, thereby enhancing overall team productivity and project delivery timelines. - Constructed a RAG system for accurate information delivery, decreasing response time by 35% and increasing customer satisfaction. - Built a production-grade LLMOps pipeline using Python sentence-transformers, OpenAI, LangChain, and ChromaDB; optimized deployment and reduced model iteration times by 60%			
Software Engineer - Data Scientist	Cognizant	Pune, INDIA	06/2021 - 01/2024
- Implemented a targeting model generating \$1M+ in healthcare revenue with 90% accuracy using anomaly detection. - Enhanced accuracy by 5-10% over traditional models with advanced supervised learning and Neural Networks.. - Reduced false alarms in fraud detection by 20% using Logistic Regression and Random Forests; received the 'Innovation Award' - Cut administrative overhead by 40% with an AI-driven prior authorization system.. - Created a sophisticated chatbot utilizing Gemini and Vector DB, achieving an impressive 90% accuracy in understanding user queries; the tool is now uses by over 15 team members for efficient data retrieval. - Engineered an advanced information retrieval chatbot that seamlessly extracts and processes data from PDF, DOCX, and TXT formats; integrated cutting-edge language models to enhance natural language understanding, boosting query response accuracy by 40%. - Tech Stack Python, Django, ML/DL frameworks, MongoDB, MLFlow, GCP, LangChain, Faiss, and Git for CI/CD pipeline.			
Programmer Analyst	Conizant	Chennai, INDIA	12/2019 - 06/2021
- Built data visualization dashboards to track key performance indicators, increasing reporting efficiency by 30%. - Integrated data from over 10 databases, APIs, and external sources, improving data consistency by 25%. - Collaborated with teams to align data efforts with business goals, enhancing data-driven decision-making across 5 departments. - Assisted in developing and training machine learning models, improving prediction accuracy by 20% under senior guidance.			

Education

Master of Technology in Data Science & Engineering	BITS Pilani	Pilani, India	09/2021 - 10/2023 CGPA: 7.9/10
Bachelor of Engineering in Computer Science & Engineering	Oriental Institute of Science & Technology	Bhopal, India	

Certifications and Awards

- First Prize : Secured top prize at the Trizetto Hackathon for a patient data integration approach, earning 95% positive feedback. - OpenAI Whisper Hackathon Participant: Created an app to assist individuals with hearing disabilities, enhancing their quality of life - GCP Professional ML Engineer: Certified in Google Cloud Platform (02/2023). - Certified MLOps Developer: Dataiku certification for ML model operations. - IBM Certification: IBM professional data science certification			
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Accident/Fall Detection using Yolo Algorithms:

- Worked on real-time human fall detection system using Python, YOLO, and OpenCV with CUDA toolkit, achieving 95% accuracy in fall detection.
- Designed a smart camera surveillance system for highway accident detection using CNN and YOLO, automating ambulance dispatch and reducing emergency response time by 40% through Python's SMTP library

Potato Disease Classification

- Developed a mobile app for potato disease classification, achieving 92% accuracy in detecting Early Blight and Late Blight.
- Quantized the CNN model with TensorFlow Lite, reducing model size by 80% and improving inference speed by 50%.
- Utilized Google Cloud Functions for the system, achieving a 25% reduction in cloud costs, and integrated TensorFlow Serving with FastAPI for real-time predictions on ReactJS and React Native interfaces.
- **Technologies: TensorFlow, CNN, TF Lite, Data Augmentation, TF Serving, FastAPI, ReactJS, React Native, Google Cloud Functions.**

Text Summarization Using BERT

- Applied BERT and NLP toolkits to summarize chat context, improving agent efficiency by 35% and enhancing customer service response times by 25%.
- Improved agent efficiency by providing quick access to previous chat history for better customer support.

US VISA Prediction

- Applied production grade code to predict structure data. Provide database connectivity with **MongoDB**. EDA and feature engineering
- Explored various ML models to achieve optimal results through gridSearchCV and hyperparameter tuning, and containerized the solution with Docker for AWS integration.

MLOps Project on AWS for Topic Modeling

- Established an end-to-end machine learning pipeline for topic modeling on AWS, cutting processing time by 40%.
- Dockerized and deployed a Python-based LDA topic modeling application on AWS ECS, cutting deployment time by 60% through a blue/green deployment strategy.
- Automated CI/CD pipelines, reducing manual effort by 70% using AWS CodeCommit, CodeBuild, CodeDeploy, and CodePipeline, while ensuring high availability with EC2 and ECS load balancing.